

CONNECTIVITY VERSUS SECURITY:

Design Engineers' Dilemma

How to design hardware to offer maximum connectivity without sacrificing security? That is the question I attempt to answer while explaining this challenge from the perspective of designing automobiles...



SRIVIDYA SUNDAR,
Consultant -
Architect for
Connectivity and
Smart Systems,
Ather Energy
(Individual Tech
Consultant)

Recent reports of vehicle thefts in Europe have highlighted vulnerabilities in smart key systems, which were intended to enhance security. These systems relied on outdated wireless tech, creating security gaps. In response, the Bluetooth SIG introduced Channel Sounding in the new Bluetooth specifications, improving precision and security for proximity-based systems, such as keyless entry.

The first real test, however, was whether Channel Sounding could become a viable product. Moving beyond technical validation, key questions emerged: What do customers need? Will it perform reliably across scenarios? Can it be low-cost, scalable, and compatible with existing systems?

Let's explore this through the vehicle

access use case. Customers expect a seamless and secure experience when accessing their vehicle, just as they do with a mechanical key. That means:

- Consistency: It should work the same way every time.
- Reliability: No glitches. No waiting.
- Security: It must be tamper-proof.
- Affordability: It should not drive up the vehicle's cost.

The Bluetooth Channel Sounding feature checks many boxes—accuracy, security, and theoretical reliability.

But in the real world, the picture is more complex. Most smart keys are integrated into smartphones, raising the question: Is Channel Sounding supported on phones? The answer was no at the time, and widespread adoption might take several years to occur. It is important to understand the technical complexity (software update vs hardware design) and the market conditions that factor into the adoption of any new standard.

Another challenge is latency. Cars can pair with multiple phones using Bluetooth for audio, fitness, location, and other purposes, making a low-latency connection difficult. While Channel Sounding interleaves ranging and data exchange, its performance in noisy environments with multiple BLE devices is yet to be proven at this time. Great on paper, but uncertain in real-world product terms.

When selecting an architecture and technology for a product with connectivity, several key considerations should be taken into account. Often, thinking about